# Evaluating and Extending Mean Flows for One-Step Generative Modeling With Resource Constraints

**Joar Paganus** **Haoran Jin** **Gawtam Chithra Ramesh**

## Abstract

MeanFlow is a recently proposed framework for one-step generative modeling that learns an average velocity field over finite time intervals, enabling efficient sampling without iterative refinement. In this work, we reproduce key results from the original MeanFlow paper on MNIST and CIFAR-10 and conduct additional ablation studies on time sampling and time conditioning. We further analyze the suitability of Classification Accuracy Score (CAS) as an evaluation metric for conditional generative models and demonstrate that it becomes unreliable under guidance-based sampling. Our results provide practical insight into MeanFlow training dynamics.

## 1 Introduction

Flow Matching and diffusion models have seen widespread adoption in modern generative modeling since their introduction. However, a key limitation of these methods is their reliance on iterative sampling during generation, which incurs high computational cost. This challenge has led to growing interest in few-step and even one-step generative models that aim to retain comparable sample quality while substantially reducing sampling cost. Recently, MeanFlow models were proposed by [1], a framework for one-step generative modeling. MeanFlow directly models a mean velocity field between two time points, enabling a single evaluation of the learned function to perform the entire generative transformation.

In this work, we focus on:

- **[Reproducibility study] Reproduce the results from the original study.** We verify the implementation and results from [1] by conducting similar experiments on CIFAR-10 and MNIST.
- **[Extended work]** We perform additional ablation studies on the time sampler, time conditioning, and Classifier-free Guidance (CFG) parameters not included in the original paper and provide further understanding of why MeanFlows work well by comparing to decomposed trivial baselines. We also evaluate our MeanFlow model with an alternative evaluation metric for conditional generative models, Classification Accuracy Score (CAS) [2] and show why the metric should not be used for evaluation of CFG parameters.

We find that many of the MeanFlow parameters are dataset dependent and tuning of these is important if optimal generation quality is to be achieved using MeanFlow.

We implement the complete training and evaluation pipeline in PyTorch, following the experimental protocol of the original paper. Our implementation and configuration files are publicly available at: `https://github.com/joarp/meanflow-reproducibility`

Project report for DD2610

## 2 Related Work

Diffusion models and Flow Matching provide a principled framework for generative modeling by learning continuous transformations from simple priors to complex data distributions [3, 4, 5]. While Flow Matching avoids explicit score estimation and unifies diffusion models with continuous normalizing flows, both paradigms typically rely on iterative numerical solvers at inference time, resulting in high sampling cost.

This limitation has motivated the development of few-step and one-step generative models. Many approaches achieve this via distillation, compressing pretrained diffusion models into a small number of steps or a single step. Although effective, distillation-based methods introduce additional training stages and dependencies on large teacher models, which complicates reproducibility and accessibility. Standalone one-step methods instead aim to learn feedforward generators directly, without relying on pretrained teachers. Consistency Models [6] and related extensions enforce agreement between network predictions across time, but often depend on carefully designed discretization schedules and auxiliary constraints.

MeanFlow [7] differs from these approaches by introducing an explicit, physically motivated training target: the average velocity over a finite time interval. This formulation naturally leads to a two-time objective that recovers standard Flow Matching in the limit of vanishing interval length, while enabling one-step generation when evaluated over the full interval. Unlike methods that impose self-consistency through auxiliary losses, MeanFlow's two-time structure follows directly from a deterministic identity of the underlying flow.

Several recent works also explore two-time or large-step formulations, such as Shortcut Models [8] and Inductive Moment Matching [9]. While these methods improve sampling efficiency, they typically rely on additional losses or constraints whose relationship to the underlying continuous dynamics is indirect. In contrast, MeanFlow's formulation remains closely tied to the geometry of the learned flow, a property we explicitly probe in our experiments through simplified baseline decompositions. Furthermore, Classifier-free Guidance (CFG) is commonly used to improve conditional generation quality in diffusion and flow-based models [10]. MeanFlow incorporates guidance directly into the learned average-velocity field, enabling guided one-step sampling without additional inference cost.

## 3 Data

In [1], MeanFlow is trained on ImageNet 256×256 and unconditionally on CIFAR-10. Due to limited computing resources, we use CIFAR-10 [11] and MNIST [12].

## 4 Method

### 4.1 MeanFlow Models

In MeanFlow models, a new field representing the average velocity is introduced. Denoting $v$ as the instantaneous velocity used in standard Flow Matching, the average velocity $u$ from time step $r$ to time step $t$ is defined as:

$$u(z_t, r, t) \coloneqq \frac{1}{t-r} \int_r^t v(z_\tau, \tau) d\tau. \tag{1}$$

Note that letting $r \to t$, the average velocity reduces to the instantaneous velocity. Similarly to in the original paper, we perform training by parameterizing a network $u_\theta$ and minimizing the following objective:

$$\mathcal{L}(\theta) = \mathbb{E}||u_\theta(z_t, r, t) - \text{sg}(u_{\text{tgt}})||_2^2, \tag{2}$$

$$\text{where} \quad u_{\text{tgt}} = v(z_t, t) - (t-r)(v(z_t, t)\partial_z u_\theta + \partial_t u_\theta), \tag{3}$$

and sg refers to the stop-gradient operation. For a full derivation of the objective, we refer to the original paper [1]. MeanFlow recovers standard Flow Matching in the limit $r \to t$, while extending it to finite intervals by supervising an average velocity field. The ratio of samples with $r \neq t$ therefore controls how often the model is trained on instantaneous Flow Matching targets versus finite-interval MeanFlow targets, effectively interpolating between the two regimes during training.

### 4.2 Classifier-free Guidance

In the original paper, two versions of Classifier-free Guidance (CFG) are proposed. In this paper, we use the improved formulation from eq (21) in the appendix of [1], which is equivalent to the non-improved version if $\kappa = 0$. Guidance is introduced through a modified target velocity that linearly combines the sample-conditional velocity with both class-conditional and class-unconditional network predictions:

$$\tilde{v}_t = \omega \underbrace{(\epsilon - x)}_{\text{sample } v_t} + \kappa \underbrace{u_\theta^{\text{cfg}}(z_t, t, t \mid \mathbf{c})}_{\text{class-cond output}} + (1 - \omega - \kappa) \underbrace{u_\theta^{\text{cfg}}(z_t, t, t)}_{\text{class-uncond output}} . \tag{4}$$

The parameter $\omega$ controls the overall guidance strength, while $\kappa$ mixes conditional and unconditional network outputs in the training target. This formulation is equivalent to the standard (non-improved) version of CFG with an effective guidance scale $\omega' = \omega/(1 - \kappa)$, while additionally exposing the model to both conditional and unconditional predictions during training. The objective then becomes:

$$\mathcal{L}(\theta) = \mathbb{E}||u_\theta^{\text{cfg}}(z_t, r, t \mid \mathbf{c}) - \text{sg}(u_{\text{tgt}})||_2^2, \tag{5}$$

$$\text{where} \quad u_{\text{tgt}} = \tilde{v}_t - (t - r)(\tilde{v}_t \partial_z u_\theta^{\text{cfg}} + \partial_t u_\theta^{\text{cfg}}). \tag{6}$$

### 4.3 Evaluation

#### 4.3.1 MNIST - Classification Accuracy Score

For MNIST, commonly used metrics such as Fréchet Inception Distance (FID) are not well-suited, as MNIST images are low-dimensional and lack the semantic complexity assumed by feature-based perceptual metrics. Furthermore, there is no widely accepted quantitative metric in the literature for evaluating the quality of synthesized MNIST samples. One proposed method for evaluating conditional generative models that is applicable on MNIST is Classifier Accuracy Score (CAS) [2]. In their work, CAS is used to evaluate generative models by training classifiers on generated data and evaluating them on real data, and is compared against FID and Inception Score (IS) on ImageNet. They find that IS and FID unfairly penalize models based on likelihood and that CAS is particularly useful in uncovering model deficiencies. We use a similar metric for our MNIST experiments by generating a synthetic MNIST training set of equal size to the original MNIST training set (60,000 samples), and train a Convolutional Neural Network (CNN) exclusively on these generated samples. The trained classifier is then evaluated on the original MNIST test set. If the classifier achieves test accuracy comparable to a classifier trained on real MNIST data, this indicates that the generated samples form a suitable surrogate for the true data distribution. The strengths of this metric include:

- Sensitivity to mode collapse, as limited diversity leads to poor generalization on the real test set.
- A high score indicates that the generated samples exhibit class-consistent structure and diversity comparable to the original MNIST dataset.

We note that this metric does not directly measure perceptual quality, but instead evaluates the usefulness of the generated data as a training distribution. Weaknesses include:

- Perceptually better-looking samples can achieve remarkably lower CAS.
- The metric does not provide valuable insight when using guidance mixing (i.e. $\omega > 1$).
- The classifier must use the same normalization as the original dataset, which can introduce additional bias when the generated distribution deviates from the true data manifold.

Therefore, we make sure to also include qualitative evaluations in our experiments. Furthermore, this study should give insight into when and how CAS can be applied. We argue that it is a useful metric when CFG parameters are fixed, but that it is insufficient as a metric for evaluation of the CFG parameters $(\omega, \kappa, p_c)$ since CAS becomes unreliable under CFG. Classifier-free guidance explicitly modifies the sampling distribution by extrapolating beyond the learned data manifold in order to increase class separability. While this often improves perceptual quality, it also introduces distributional shifts which CAS is sensitive to.

The CNN used for CAS evaluation is a shallow two-conv, two-FC architecture which achieves a test accuracy of $99.10\%$ on the MNIST testset after training on the original training set. We train for $5$ epochs and use an ADAM optimizer with $\beta_1 = 0.9,\ \ \beta_2 = 0.95$ [13], and learning rate $0.001$. The same parameters are used for all evaluations throughout the paper.

#### 4.3.2 CIFAR10

For CIFAR-10 we follow common practice and use Fréchet Inception Distance (FID) and Inception Score (IS) for quantitative evaluation.

## 5 Experiments

### 5.1 MNIST Ablation Studies

**Experimental Setting.** For the MNIST experiments, we use a lighter version of the U-NET [14] architecture with: model_channels=32, channel_mult=[1,2], num_blocks=1, where we turn off attention and dropout. In all MNIST experiment we set number of epochs to $10$, learning rate to $0.001$, ema decay to $0.9995$, use a linear scheduler, and an ADAM optimizer with $\beta_1 = 0.9,\ \beta_2 = 0.95$ [13]. We perform ablation studies on the other training parameters, where, unless mentioned otherwise, we use the default setting in table 1. All experiments use 1-NFE unless stated otherwise.

| **Parameter** | time sampler | scheduler | $r \neq t$ | $\omega$ | $\kappa$ | $p_c$ (cfg_drop_ratio) |
|---|---|---|---|---|---|---|
| **Value** | logit normal, $(\mu = -2, \sigma = 2)$ | linear | 0.75 | 3.0 | 0.0 | 0.0 |

Table 1: Default parameters used for MNIST experiments unless stated otherwise. Note that we test multiple different combinations of the above parameters in the ablation studies below.

**Ratio of sampling.** We study the effect of the ratio of sampling with $r \neq t$ by performing similar experiments as in the original paper. We find that a ratio of $75\%$ works best for MNIST considering both CAS in Table 2a and qualitative samples in Figure 1a. For lower-dimensional datasets such as MNIST, we hypothesize that a larger ratio of samples with $r \neq t$ is beneficial because the underlying data manifold is substantially simpler than in high-resolution natural images. As a result, the average velocity field can be learned more directly without requiring frequent reinforcement of the instantaneous velocity field. This reasoning is consistent with the observation that the original authors adopted a higher ratio ($75\%$) when training on CIFAR-10, despite reporting a lower optimal ratio ($25\%$) for ImageNet 256×256. In Figure 1a, we observe that when using a ratio of $0\%$ and the model reduces to standard flow matching under CFG, the sample quality is much lower than with MeanFlow enabled.

**Time sampler.** The original paper uses different logit-normal (lognorm) time-sampling parameters depending on the dataset: lognorm(-2.0, 2.0) for CIFAR-10 and lognorm(-0.4, 1.0) in the ImageNet-256×256 ablation. We test a Uniform sampler and then focus our analysis on the $\mu$ parameter of the logit-normal sampler. Decreasing $\mu$ pushes samples toward smaller $t$ (and therefore smaller $r$ after sorting), while increasing $\mu$ biases samples toward larger $t$. For MeanFlow, samplers with more probability mass at small times place more training signal close to the data manifold. To study how this interacts with multi-step generation, we additionally evaluate 4-NFE sampling. The results are shown in Table 2b. We observe that samplers most concentrated near small times (lognorm(-3,2) and lognorm(-2,2)) achieve higher CAS when moving from 1-NFE to 4-NFE, whereas the Uniform and lognorm(-1,2) samplers show little change. Although our initial intuition was that the Uniform sampler would benefit from 4-NFE sampling, the results instead suggest that sufficient training signal close to the data manifold is more important. The observed improvement in CAS for two of the samplers agrees with the original paper, where increasing the number of NFEs from 1 to 2 led to improved FID on ImageNet. At the same time, we find that concentrating the time sampler too strongly (as in lognorm(-3,2)) reduces CAS relative to lognorm(-2,2), indicating a trade-off between emphasizing small times and maintaining sufficient coverage over the time interval.

**CFG combinations.** While we wanted to include rigorous CFG ablations, our choice of evaluation metric (CAS) did not allow for the rigour we wanted. We empirically show the limitations of CAS as

| % of $r \neq t$ | CAS |
|---|---|
| 0% (=FM) | 68.82% |
| 25% | 84.45% |
| 50% | 86.17% |
| 75% | **87.43%** |

(a) Ratio of sampling. Experiments used $\kappa = p_c = 0$.

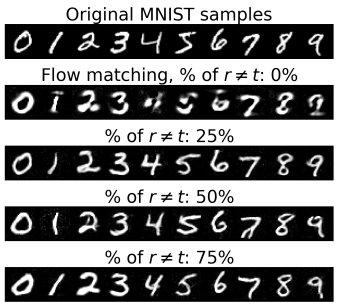
| $t, r$ sampler | 1CAS | 4CAS |
|---|---|---|
| uniform(0,1) | 84.35% | 83.97% |
| lognorm(-2, 2) | **87.43%** | **88.66%** |
| lognorm(-1, 2) | 84.04% | 84.21% |
| lognorm(-3, 2) | 85.40% | 87.39% |

(b) Different $t$, $r$ samplers and parameters. Experiments used $\kappa = p_c = 0$. 1CAS indicates 1-NFE CAS and 4CAS indicates 4-NFE CAS.

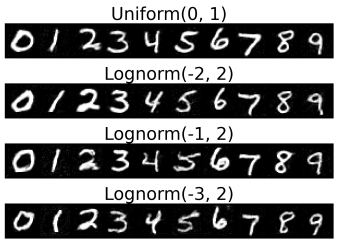
| $\omega$ | CAS |
|---|---|
| 1.0 | **96.12%** |
| 1.5 | 93.58% |
| 2.0 | 90.42% |
| 3.0 | 87.43% |
| 5.0 | 72.53% |

(c) Different $\omega$ values. Experiments used $\kappa = p_c = 0$.

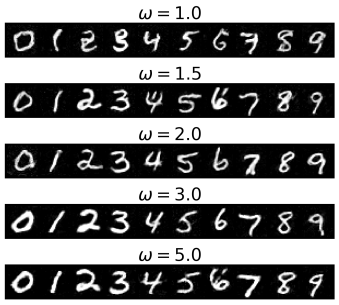
Table 2: CAS for different ablations.




(a) Samples for different $r \neq t$.




(b) Samples for different time samplers used during training (1-NFE).




(c) Samples for different $\omega$.

Figure 1: Corresponding samples to ablation studies in table 2.

a metric when using CFG by ablations on $\omega$ where Table 2c reports CAS and Figure 1c, 2a, and 2b shows qualitative samples using different $\omega$. We observe that the CAS is higher without guidance mixing ($\omega = 1.0$), but the generated samples are of higher quality with $\omega = 3.0$. The qualitative difference between $\omega = 1.0$ and $\omega = 3.0$ is especially evident in Figure 2a and 2b, where using $\omega = 1.0$ yield multiple patched and inconsistent samples. This ties nicely to our discussion in the Classifier Accuracy Score section about CAS not being well-suited as an evaluation metric when varying CFG parameters. Note that training without guidance mixing still means we train with conditioning, just without guidance mixing as per Eq. 4 and 6.

In Figure 2c, samples from MeanFlow models trained with different values of the conditioning drop probability $p_c$ are shown. We observe that introducing $p_c > 0$ consistently leads to unwanted white blur or noise in the background regions of generated samples. We hypothesize that this behavior is a consequence of MNIST being a low-dimensional dataset where most pixels in the background are close to zero. When conditioning is dropped during training, the model is encouraged to interpolate between conditional and unconditional behaviors. While this may be beneficial for complex datasets, it here introduces additional variance in regions that should remain nearly deterministic, resulting in visible background artifacts. No ablation studies were performed on $p_c$ in the original paper.

### 5.2 CIFAR-10 Results

**Experimental setting.** We trained an unconditional MeanFlow model on CIFAR-10 (32×32) using a Song-style U-Net with model_channels=128, channel_mult=[1,2,2,2], two residual blocks per resolution, and attention at resolutions 16 and 8. Inputs were normalized to $[-1, 1]$. Training employed a linear interpolation schedule $z_t = (1 - t)x + t\epsilon$ and the MeanFlow objective with mixed sampling, where $r \neq t$ with probability 0.75 and $r = t$ otherwise. Optimization was performed using Adam with a learning rate of $3 \cdot 10^{-4}$ and EMA decay 0.9995.

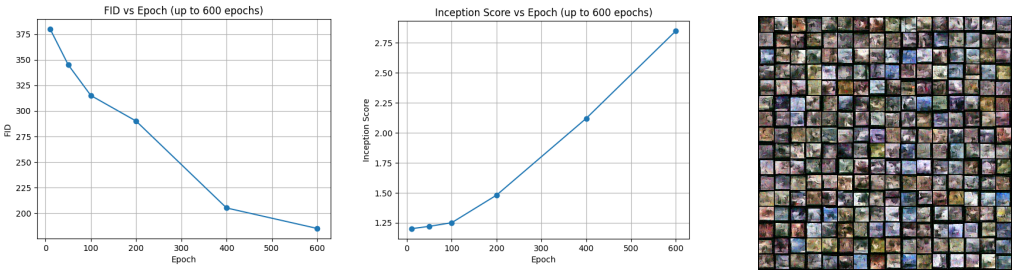
Due to limited computational resources, training was carried out with a batch size of 32 and terminated after approximately 600 epochs (corresponding to roughly 60k optimization steps), after which further training was not feasible.

(a) Samples when using $\omega = 1.0$ (b) Samples when using $\omega = 3.0$ (c) Samples for different $p_c$.

Figure 2: Samples highlighting limitations of CAS as a quantitative metric for generative quality in (a) and (b). Samples using different $p_c$ values in (c).

**Quantitative results.** Figure 3a reports the Fréchet Inception Distance (FID) computed on 50k generated samples using one-step (1-NFE) sampling with EMA weights. FID decreases rapidly during early training and reaches its best observed value of approximately 160 around epoch 600, at which point training was stopped. Figure 3b shows the corresponding Inception Score (IS), which increases monotonically throughout training and saturates near the final checkpoint.

The obtained FID values are substantially worse than the reported values in the original MeanFlow study. This discrepancy is expected given the large mismatch in training scale: the original work trains with batch size 1024 for several hundred thousand optimization steps, whereas our experiments use a batch size of 32 and terminate after roughly 60k steps. For one-step generative models in particular, large batch sizes and long training horizons are critical for stable convergence and strong quantitative performance.




(a) FID vs epoch (b) Inception Score vs epoch (c) Samples at epoch 600

Figure 3: CIFAR-10 MeanFlow results with one-step sampling. Left: FID over training. Middle: Inception Score over training. Right: samples at epoch 600 corresponding to best FID.

**Qualitative results.** Figure 3c shows samples generated at epoch 600 using one-step sampling. The model produces diverse color distributions and textured patterns, and in some cases captures coarse object-like structure. However, samples remain blurry and often lack global semantic coherence, consistent with the relatively high FID and the limited training budget. These qualitative observations align with the quantitative metrics and further highlight the importance of large-scale training for MeanFlow on CIFAR-10.

### 5.3 MNIST Trivial Baselines

We compare our MeanFlow model to two trivial baselines:

(T1) Directly predict $x_r$ with $x_t$.

(T2) Always set the terminal time to be 0 ($r = 0$) during training.

The baselines are designed to isolate the contributions of flow-based modeling and explicit time-interval conditioning. (T1) removes the underlying velocity field and performs direct end-to-end regression, testing whether the geometric structure induced by flow matching is essential. (T2) collapses the two-time MeanFlow formulation into a single-time objective, probing whether conditioning on variable time intervals $(r, t)$ is necessary. Figure 4 shows samples from (T1), (T2), and the default MeanFlow model, while Table 3 reports CAS. The relatively small performance gap of only $2\%$ in CAS for (T2) compared to the default MeanFlow model suggests that a substantial part of MeanFlow's benefit comes from learning an average velocity over a finite interval, while conditioning on multiple $(r, t)$ pairs might primarily act as a form of regularization rather than being critical for one-step performance. In contrast, the results for (T1) highlight the necessity of flow-based modeling. By discarding the velocity field and directly regressing endpoints, (T1) fails to capture the underlying transport geometry learned by flow matching, leading to substantially worse performance.

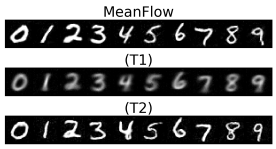



Figure 4: Samples from trivial baselines.

Table 3: Trivial baseline results where the MeanFlow model used the default settings in Table 1.

| Method | CAS |
|---|---|
| MeanFlow | **87.43%** |
| T1 | 76.34% |
| T2 | 85.26% |

## 6 Challenges

Since none of the group members had access to multiple GPU:s on GCP we had to resort to a laptop GPU and made it possible to get one timeslot where we had access to a cluster for the CIFAR-10 experiment. This made our inital plans of conducting all experiments on CIFAR-10 non feasible and that was the reason we used MNIST for most of the experiments. Furthermore, we struggled to find valuable quantitative metrics for the evaluation on MNIST. The exact UNet architecture (e.g. resample_filter, channel_mult, etc) for the original experiments on CIFAR-10 was not shown in the paper and consultation of the source code was needed to use the same architecture. Apart from that, we found the original paper to be transparent about parameter settings and implementation details.

## 7 Conclusion

In this work, we reproduced key results of the MeanFlow framework and conducted additional analyses to better understand its training dynamics and evaluation. Our ablation studies highlight the role of the time sampler, classifier-free guidance parameters, and ratio of sampling $r \neq t$, while comparisons to trivial baselines emphasize the importance of flow-based modeling. For MeanFlow models trained on MNIST, we find that fixing the terminal time to $r = 0$ throughout training has little effect on sample quality, that introducing a nonzero CFG drop ratio leads to unwanted blur artifacts, and that using a higher proportion of samples with $r \neq t$ is beneficial. Overall, our results indicate that several parameters introduced by MeanFlow are dataset-dependent and require careful tuning to achieve optimal generation quality. Our findings support MeanFlow as an effective one-step generative approach, while clarifying practical considerations for its evaluation and use.

## 8 Ethical Consideration

MeanFlow introduces a more compute-efficient one-step generative modeling framework without changing the risks of diffusion and flow-based models. Ethical considerations are therefore inherited rather than novel, including dataset bias propagation and potential misuse of image synthesis, while the method itself relies only on standard benchmarks (ImageNet, CIFAR-10) and involves no human subjects or new data collection. The experimental design is transparent and reproducible, although training large-scale MeanFlow models remains computationally and energy intensive, particularly for

high-resolution datasets such as ImageNet. Our reproduction and extensions remain within the same ethical scope, using MNIST and CIFAR-10 only, and add an explicit methodological contribution by demonstrating that Classification Accuracy Score can be misleading under classifier-free guidance, reducing the risk of incorrect evaluation and overstated claims. Societally, MeanFlow's primary impact is reduced inference cost and energy usage through one-step sampling, lowering barriers to research and deployment while marginally lowering the cost of misuse common to generative models. In terms of UN SDGs, the work most directly supports SDG 9 through more efficient AI methods, and SDG 12 and SDG 13 by reducing computational and energy requirements.

## 9 Limitations

We support the experiments conducted and the original MeanFlow paper as a whole, but have identified that optimal generation quality can only be achieved after performing a significant number of training runs to evaluate the different possible parameter choices.

## 10 Self Assessment

We believe our project should receive a A for the following reasons:

- Except for the model architecture (U-NET), we reimplement the proposed method in the MeanFlow paper from scratch, following only the paper.
- We demonstrate a strong understanding of the strengths and limitations of different evaluation methods for generative models. In particular, we provide insight into shortcomings of the Classifier Accuracy Score (CAS) when used in conjunction with classifier-free guidance.
- We perform interesting ablations not included in the original paper related to the selection of: time sampler, time conditioning, and cfg_drop_ratio parameter.
- We reflect on decompositions of the proposed MeanFlow method through comparisons to trivial baselines, providing clearer insight into the role of flow-based modeling and the impact of removing variability in the terminal time by fixing $r = 0$ during training.

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